Visualization Development Phase-V

Business Question and Visualization Report

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Defining the 9 Core Business Questions & Visualizations

This report establishes the 9 core analytical queries derived from our 3 personas (Rajan, Divya, Arvind) mapped to specific visual implementations in Tableau across the 6,464 race records.

No.	Business Question / Goal	Target Scenario	Visual Type	Measures & Dimensions
V-01	How do mid-tier team championship points benchmark against Red Bull & Mercedes across 17 seasons?	Scenario 1 (Strategist)	Multi-Line Chart	Cols: season | Rows: SUM(points) | Color: team
V-02	How does tyre compound degradation vary across Permanent, Street, and High-Speed circuits?	Scenario 1 (Strategist)	Heatmap / Highlight Table	Cols: circuit_type | Rows: starting_compound | Color: AVG(points)
V-03	Which nations represent the strongest global driver recruitment talent pools by points scored?	Scenario 1 (Strategist)	World Map / Treemap	Geo: country | Size: SUM(points) | Tooltip: driver
V-04	Who are the top all-time drivers by cumulative points, wins, and podiums for broadcast telecasts?	Scenario 2 (Broadcast)	Horizontal Bar Chart	Rows: driver | Cols: SUM(points) | Sort: Desc | Color: team
V-05	Which host countries have hosted the highest volume of Grand Prix rounds since 2010?	Scenario 2 (Broadcast)	Ranked Bar Chart	Rows: country | Cols: COUNTD(race_id) | Sort: Desc
V-06	What is the distribution of podium appearances across constructors and drivers?	Scenario 2 (Broadcast)	Stacked Bar / Donut	Cols: team | Rows: SUM(Is_Podium) | Color: driver
V-07	What failure modes (Engine, Collision, Gearbox, Brakes) dominated each technical regulation era?	Scenario 3 (Compliance)	100% Stacked Bar	Cols: season / Era | Rows: SUM(dnf) | Color: dnf_reason
V-08	Has overall grid finish reliability improved from the V8 era to the Ground Effect era?	Scenario 3 (Compliance)	Trend Line Chart	Cols: season | Rows: Reliability_Rate (%)
V-09	What is the career finish vs. retirement ratio for active and past drivers?	Scenario 3 (Compliance)	Stacked Diverging Bar	Rows: driver | Cols: COUNT(race_id) & SUM(dnf)